

An Emerging Class Of GRAPHENE BASED ELECTRONICS



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In an era of electronics based civilisation there is always a demand for more, faster, better and cheaper electronic devices. Electronics started with copper wires and moved to semiconductors (silicon and germanium) base, but now scientists and engineers are putting in efforts to design and develop graphene based electronics for which new applications that fulfil all qualities of desired electronics are emerging very fast. These electronics meet the demands of electronic device-happy and data-centred world of consumers.

Meeting these demands requires technologies for processing and storing information. Now, with the use of graphene, a significant obstacle to the development of next-generation device technologies appears to have been overcome. In simple terms, graphene is a thin layer of pure carbon. It is a single, tightly-packed layer of carbon atoms that are bonded together in a hexagonal honeycomb lattice. Graphene is the world's first 2D material. Since its isolation in 2004 it has captured the attention of scientists, researchers and industry worldwide with its unique properties.

Some properties of graphene are:

1. It is ultra-light yet immensely tough.
2. It is 200 times stronger than steel, but is incredibly flexible.
3. It is the thinnest material possible and is transparent.
4. It is a superb conductor and can act as a perfect barrier; not even helium can pass through it.

Graphene has been proven to be much more efficient at conducting electrons than silicon, and is also able to transfer electrons at much faster speeds (relatively speaking, 1000 kilometres per second, which is 30 times faster than silicon). In the next few years you will begin to see products from consumer electronics companies using graphene, and electronic devices based on flexible, robust, touch-screen devices such as mobile smart-phones and wrist watches.

This could mean foldable televisions and telephones, and eventually electronic flexible newspapers and publications that can be updated via wireless data transfer. Being extremely translucent, in the coming years you can also expect to be able to fit intelligent (and extremely robust) windows in your home, with (potentially) virtual curtains or projected images of your choice.

Soon you will begin to see clothing containing graphene-enhanced photovoltaic cells and super capacitors. So you will be able to charge your mobile telephones and tablet computers in a matter of minutes (even seconds) while walking to work. You may possibly even see security-orientated clothing offering protection against unwanted contact with the use of electrical discharge.

Possibilities of what we can achieve with the materials and knowledge we possess, have been blown wide open. It is now conceivable to imagine such amazing

